Why did Medicare ACOs reset their benchmarks early? Results

This notebook reports the results of the pre-registered study (OSF registration c53w9, DOI 10.17605/OSF.IO/C53W9; plan and code at commit `a4064f5`). It only reads the pipeline's results files in `analysis/mssp-benchmark-ratchet/results/` and recomputes nothing. The blog post is the authoritative report; the verdicts below use the same wording as its dashboard.

Units: "pp" is percentage points. Benchmark components are in percentage points of the ACO's benchmark. Intervals are 95% equal-tailed intervals; p-values are two-sided.

In [ ]:
```
import hashlib
import json
import sys

import marimo as mo
import matplotlib.pyplot as plt

sys.path.insert(0, str(mo.notebook_dir()))
import dashboard_data
```

In [ ]:
```
results_dir = dashboard_data.DEFAULT_RESULTS
data = dashboard_data.build(results_dir)
raw = dashboard_data.load(results_dir)
```

In [ ]:
```
colors = {"ink": "#1f2933", "muted": "#5f6b7a", "grid": "#d9dee5", "mark":
"#2b5c9e"}

def num(x, digits=1):
    if x is None:
        return ""
    text = f"{x:.{digits}f}"
    return text.lstrip("-") if float(text) == 0 else text.replace("-",
"\u2212")

def pval(p):
    return "" if p is None else "< 0.001" if p < 0.001 else f"{p:.3f}"

def interval(r, digits=1):
    return "" if r.get("lo") is None else f"{num(r['lo'], digits)} to
{num(r['hi'], digits)}"

def table(header, rows):
    lines = ["| " + " | ".join(header) + " |", "|" + "---|" * len(header)]
    lines += ["| " + " | ".join(str(c) for c in row) + " |" for row in
rows]
    return "\n".join(lines)
```

In [ ]:
```
def forest(ax, rows, xlabel, band=None):
    """One interval per row, top to bottom, with a zero line; band shades a prior's 95% range."""
    ys = list(range(len(rows)))[::-1]
    if band:
        ax.axvspan(-band, band, color=colors["grid"], alpha=0.5, lw=0)
    ax.axvline(0, color=colors["muted"], lw=1)
    for y, r in zip(ys, rows, strict=True):
        if r.get("lo") is not None:
            ax.plot([r["lo"], r["hi"]], [y, y], color=colors["mark"], lw=2, solid_capstyle="round")
        ax.plot(r["mean"], y, "o", color=colors["mark"], ms=6, mec="white", mew=1.5)
    ax.set_yticks(ys, [r["label"] for r in rows], color=colors["ink"])
    ax.set_xlabel(xlabel, color=colors["ink"])
    ax.tick_params(colors=colors["muted"], length=0)
    ax.grid(axis="x", color=colors["grid"], lw=0.8)
    ax.set_axisbelow(True)
    for side in ("top", "right", "left"):
        ax.spines[side].set_visible(False)
    ax.spines["bottom"].set_color(colors["grid"])
    ax.set_ylim(-0.6, len(rows) - 0.4)

def figure(n_rows, height_per_row=0.45):
    fig, ax = plt.subplots(figsize=(7.5, 0.8 + height_per_row * n_rows), layout="constrained")
    return fig, ax
```

Provenance

Results are from the real CMS data. Each stage records the commit it ran at; all of them contain the registered code unchanged (see `DEVIATIONS.md`).

Stage	Commit	sha256 of results file (first 16)
validation	`fd1920f`	`046d82a1cc40548e`
power	`fd1920f`	`9d48835d8b87aac1`
confirm	`3e5cb9b`	`35bba7e38d3dd6e1`
robustness	`1debc2c`	`66f00ad1a01e2f1c`
did	`992d13f`	`8f3a6842df52d1eb`
conventional	`acc8b0e`	`3508370b700d1d00`
exploratory	`ff6e64b`	`280bbe10b738a284`

The post's dashboard data (`data.json`) matches these results.

Validation of the benchmark approximation

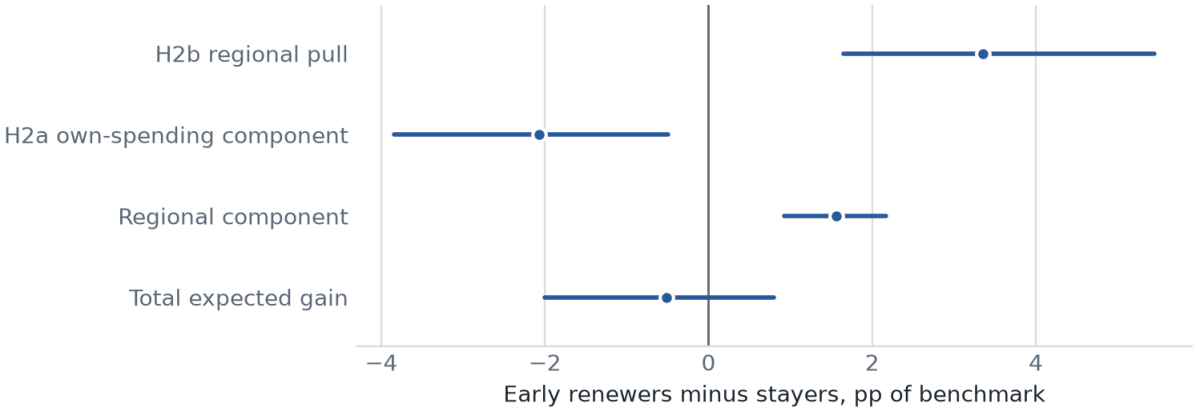
Passed: **yes**. Calibration applied: no. Measurement-error SD used in the decision model (pp of benchmark): 4.38.

Who renewed early (H2b, H2a, H1)

Early renewers minus stayers (Bayesian bootstrap; p-values from stratified permutation).

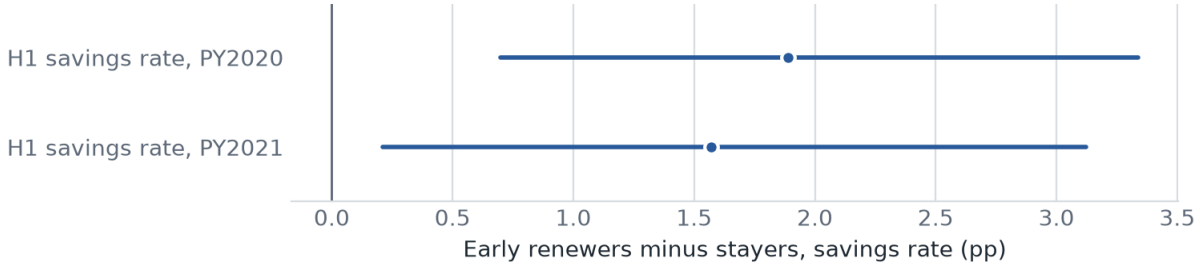
Comparison	Role	n early / stayers	Difference	95% interval	p	MDE	Verdict
H2b regional pull	primary	26 / 241	3.4	1.7 to 5.4	< 0.001	2.5	supported
H2a own-spending component	secondary	26 / 241	−2.1	−3.8 to −0.5	0.003	1.8	contradicted
H1 savings rate, PY2020	descriptive	29 / 277	1.9	0.7 to 3.3	0.010	2.7	
H1 savings rate, PY2021	descriptive	29 / 277	1.6	0.2 to 3.1	0.033	2.7	
Total expected gain	descriptive	26 / 241	−0.5	−2.0 to 0.8	0.325		
Regional component	descriptive	26 / 241	1.6	0.9 to 2.2	< 0.001		

In [ ]:
```
_c = {r["key"]: r for r in data["comparisons"]["rows"]}
_fig, _ax = figure(4)
forest(_ax, [_c[k] for k in ("H2b_pull", "H2a_own", "regional_comp",
"gain")], "Early renewers minus stayers, pp of benchmark")
_fig
```



In [ ]:
```
_c = {r["key"]: r for r in data["comparisons"]["rows"]}
_fig, _ax = figure(2)
```

```
forest(_ax, [_c[k] for k in ("H1_sav_rate_2020", "H1_sav_rate_2021")],
"Early renewers minus stayers, savings rate (pp)")
_fig
```



H3: the decision model

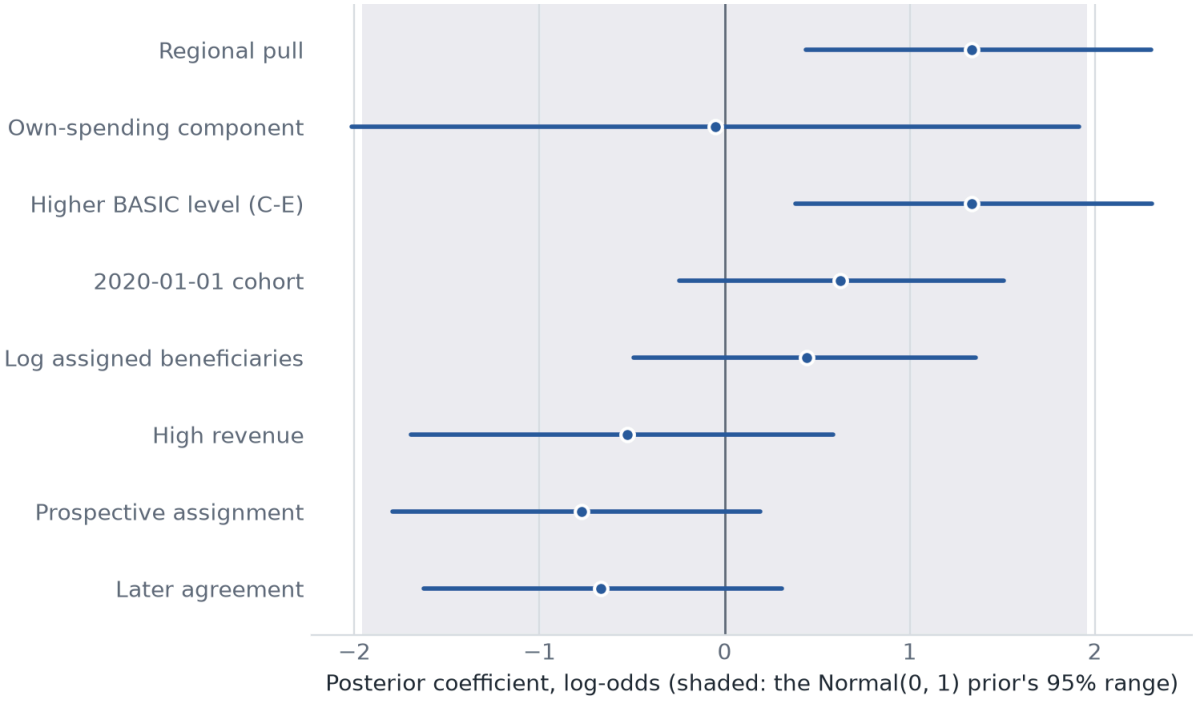
Bayesian logistic regression of early renewal, with measurement error on the own-spending component; 206 ACOs, 24 early renewers (3 early renewers and 27 stayers excluded for a missing predictor). Converged: yes. Coefficients are log-odds per 2 SD of a continuous predictor; tests are by direct simulation under the reduced model.

Coefficient	Predicted	Posterior mean	95% interval	P(> 0)	p	Decision
Regional pull	+	1.34	0.44 to 2.30	0.999	0.002	supported
Own-spending component	+	−0.05	−2.01 to 1.92	0.480	0.191	undetermined
Higher BASIC level (C-E)	+	1.34	0.38 to 2.31	0.998	0.003	supported
2020-01-01 cohort		0.63	−0.24 to 1.51	0.920		
Log assigned beneficiaries		0.45	−0.49 to 1.36	0.827		
High revenue		−0.53	−1.69 to 0.59	0.182		
Prospective assignment		−0.77	−1.79 to 0.19	0.058		
Later agreement		−0.67	−1.62 to 0.31	0.089		

Overall test (max |posterior mean| of core coefficients): observed 1.35, p = 0.004 (10,000 shuffles). Power to detect the planned effects: 17% (pull), 0% (own), 14% (BASIC level).

In [ ]:
```
_rows = data["h3"]["posterior"]["normal"]
_fig, _ax = figure(len(_rows))
forest(_ax, _rows, "Posterior coefficient, log-odds (shaded: the Normal(0,
```

```
1) prior's 95% range)", band=data["h3"]["prior_band"]["normal"])
_fig
```



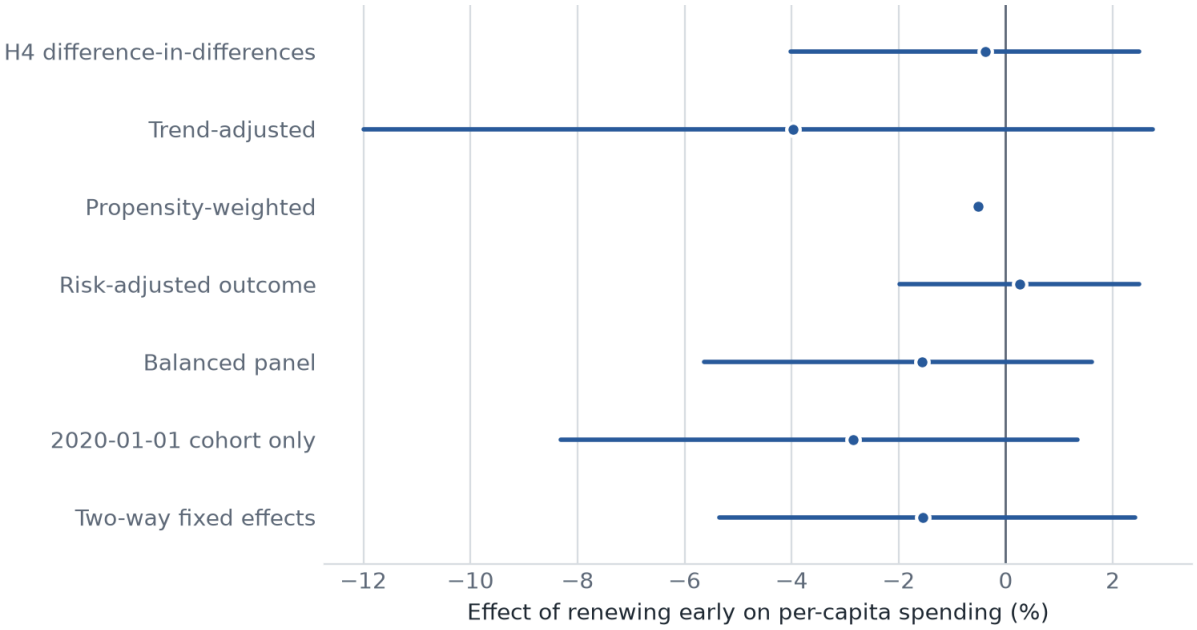
H4: what renewing did to spending

Difference-in-differences in per-capita spending, in percent; 29 early renewers and 276 comparison ACOs. Verdict: **no detectable difference** (MDE 3.9%).

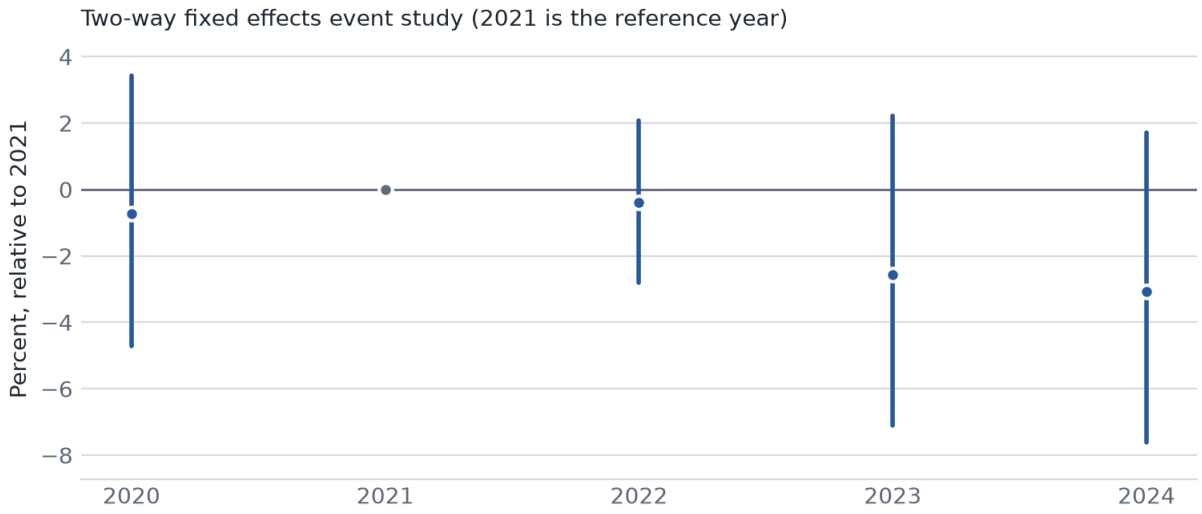
Estimate	Role	Percent	95% interval	p
H4 difference-in-differences	confirmatory	−0.4	−4.0 to 2.5	0.778
Trend-adjusted	robustness	−4.0	−12.0 to 2.7	0.142
Propensity-weighted	robustness	−0.5		0.668
Risk-adjusted outcome	robustness	0.3	−2.0 to 2.5	0.849
Balanced panel	robustness	−1.6	−5.6 to 1.6	0.298
2020-01-01 cohort only	robustness	−2.9	−8.3 to 1.3	0.130
Two-way fixed effects	conventional	−1.5	−5.4 to 2.4	0.438

Pre-trend slope difference: 0.007 (−0.023 to 0.048), p = 0.585. Equivalence condition: the pre-trend bound scaled to the post period is 0.115 against |H4| = 0.004, so it does not hold.

In [ ]:
```
_rows = data["h4"]["estimates"]
_fig, _ax = figure(len(_rows))
forest(_ax, _rows, "Effect of renewing early on per-capita spending (%)")
_fig
```



In [ ]:
```
_pts = data["h4"]["event_study"]
_fig, _ax = plt.subplots(figsize=(7.5, 3.2), layout="constrained")
_ax.axhline(0, color=colors["muted"], lw=1)
for _p in _pts:
    if not _p["reference"]:
        _ax.plot([_p["year"], _p["year"]], [_p["lo"], _p["hi"]],
color=colors["mark"], lw=2, solid_capstyle="round")
    _ax.plot(_p["year"], _p["mean"], "o", color=colors["muted"] if
_p["reference"] else colors["mark"], ms=6, mec="white", mew=1.5)
_ax.set_xticks([p["year"] for p in _pts])
_ax.margins(x=0.05, y=0.1)
_ax.set_ylabel("Percent, relative to 2021", color=colors["ink"])
_ax.set_title("Two-way fixed effects event study (2021 is the reference
year)", color=colors["ink"], fontsize=10, loc="left")
_ax.tick_params(colors=colors["muted"], length=0)
_ax.grid(axis="y", color=colors["grid"], lw=0.8)
_ax.set_axisbelow(True)
for _side in ("top", "right", "left"):
    _ax.spines[_side].set_visible(False)
_ax.spines["bottom"].set_color(colors["grid"])
_fig
```



Registered and conventional methods

Claim	Registered method	Conventional method	Agree
H2b	supported	supported	yes
H2a	contradicted	contradicted	yes
H3 regional pull	supported	supported	yes
H3 own-spending component	undetermined	undetermined	yes
H3 higher BASIC level	supported	supported	yes
H4	undetermined	undetermined	yes

The plan's agreement table (section 5.7) calls H4 undetermined unless $p < 0.05$ in a direction; that is the same verdict as "no detectable difference" in the H4 section (section 7, rule 5).

Robustness

The plan's section 5.5 items that rerun each confirmatory estimate. They do not change the decisions. H3 rows are the coefficient in each refit.

H2b regional pull (pp of benchmark)

Analysis	Estimate	95% interval	p
Confirmatory	3.4	1.7 to 5.4	< 0.001
Regression, cohort only (5.3)	2.1	0.5 to 3.8	
Regression + track, level and size (item 1)	2.2	0.7 to 3.8	
Difference in medians (item 2)	2.8	0.1 to 5.2	< 0.001
2020-01-01 cohort only (item 3)	3.7	1.8 to 5.5	0.001
Unadjusted county spending (item 9)	6.9	2.4 to 12.4	0.002

H2a own-spending component (pp of benchmark)

Analysis	Estimate	95% interval	p
Confirmatory	−2.1	−3.8 to −0.5	0.003
Regression, cohort only (5.3)	−1.5	−2.8 to −0.3	
Regression + track, level and size (item 1)	−1.3	−2.6 to 0.0	
Difference in medians (item 2)	−1.4	−3.2 to 0.0	0.085
2020-01-01 cohort only (item 3)	−1.8	−4.2 to 0.5	0.043
BY2-BY3 only in step 1 (item 6)	−1.9	−3.7 to −0.2	0.012
Complete cases (item 8)	−2.1	−3.9 to −0.5	0.005

H1 savings rate, PY2020 (pp)

Analysis	Estimate	95% interval	p
Confirmatory	1.9	0.7 to 3.3	0.010
Regression, cohort only (5.3)	1.5	0.2 to 2.9	
Regression + track, level and size (item 1)	1.2	−0.1 to 2.6	
Difference in medians (item 2)	1.4	−0.1 to 3.3	0.061
2020-01-01 cohort only (item 3)	1.8	0.5 to 3.3	0.028

Total expected gain (pp of benchmark)

Analysis	Estimate	95% interval	p
Confirmatory	−0.5	−2.0 to 0.8	0.325
Complete cases (item 8)	−0.5	−2.0 to 0.8	0.333

Regional component (pp of benchmark)

Analysis	Estimate	95% interval	p
Confirmatory	1.6	0.9 to 2.2	< 0.001
Complete cases (item 8)	1.6	0.9 to 2.2	< 0.001

Regional pull (log-odds)

Analysis	Estimate	95% interval	p
Confirmatory	1.34	0.44 to 2.30	
Firth logistic regression (item 5)	1.33	0.25 to 2.54	0.004
2020-01-01 cohort only (item 3)	1.15	−0.02 to 2.39	
BY2-BY3 only in step 1 (item 6)	1.33	0.42 to 2.26	
No measurement-error model (item 7)	1.22	0.34 to 2.20	
Complete cases (item 8)	1.34	0.38 to 2.32	
Unadjusted county spending (item 9)	0.75	−0.02 to 1.56	

Own-spending component (log-odds)

Analysis	Estimate	95% interval	p
Confirmatory	−0.05	−2.01 to 1.92	
Firth logistic regression (item 5)	−0.65	−1.74 to 0.30	0.183
2020-01-01 cohort only (item 3)	−0.03	−1.97 to 1.96	
BY2-BY3 only in step 1 (item 6)	−0.02	−1.98 to 1.97	
No measurement-error model (item 7)	−0.64	−1.51 to 0.23	
Complete cases (item 8)	−0.06	−2.04 to 1.95	
Unadjusted county spending (item 9)	−0.04	−1.99 to 1.93	

Higher BASIC level (C-E) (log-odds)

Analysis	Estimate	95% interval	p
Confirmatory	1.34	0.38 to 2.31	
Firth logistic regression (item 5)	1.55	0.53 to 2.70	0.006
2020-01-01 cohort only (item 3)	1.57	0.42 to 2.76	

Analysis	Estimate	95% interval	p
BY2-BY3 only in step 1 (item 6)	1.33	0.41 to 2.28	
No measurement-error model (item 7)	1.26	0.33 to 2.23	
Complete cases (item 8)	1.34	0.41 to 2.32	
Unadjusted county spending (item 9)	1.52	0.61 to 2.47	

Exploratory

Labeled exploratory in the plan (section 5.6): no verdicts.

H2b, early renewers minus stayers (pp of benchmark)

Measure	Estimate	95% interval	p
Regional pull (confirmatory)	3.4	1.7 to 5.4	< 0.001
Pull weighted by phase-in weight w	1.6	0.7 to 2.6	< 0.001
Capped regional component	1.6	0.9 to 2.2	< 0.001

H3 coefficient on the pull measure (log-odds)

Measure	Estimate	95% interval	p
Regional pull (confirmatory)	1.34	0.44 to 2.30	
Pull weighted by phase-in weight w	1.20	0.40 to 2.04	
Capped regional component	1.49	0.57 to 2.45	
Census region added	1.43	0.46 to 2.41	

Census region in the H3 model (log-odds)

Region	Posterior mean	95% interval
Midwest vs Northeast	0.35	−0.85 to 1.50
South vs Northeast	−0.12	−1.12 to 0.89
West vs Northeast	0.09	−1.21 to 1.38

Means by group

Variable	Early renewers	Leavers	Stayers
Savings rate, PY2020 (pp)	6.4 (n = 29)	2.7 (n = 28)	4.6 (n = 277)
Regional pull (pp of benchmark)	5.0 (n = 26)	4.8 (n = 20)	1.8 (n = 241)
Own-spending component (pp of benchmark)	−5.3 (n = 26)	−1.8 (n = 20)	−3.2 (n = 241)
Regional component (pp of benchmark)	3.7 (n = 26)	2.0 (n = 20)	2.2 (n = 241)